

MicroTex® 950

Introduction

MicroTex 950 is a PFAS-free additive that provides fine texturing and low gloss in powder coatings. The performance of MicroTex 950 can be compared to PTFE-based texturing additives such as Fluo 750TX:

Physical Property:	MicroTex 950	Fluo 750TX
Wax Type	Modified polyolefin	PTFE
Melting Point °C	135 - 140	>325
Density @ 25 °C (g/cc)	1.22	2.20
Maximum particle size (µm)	74	145

Objective

To evaluate the performance of MicroTex 950 vs. PTFE in various powder coating systems.

Test Methods

- Specular Gloss (ASTM D2457)
- Impact Resistance (ASTM D2794)
- Conical Mandrel Bend (ASTM D522)
- Crosshatch Adhesion (ASTM D3359)
- Pellet Flow (ASTM D3541)

General Formulation Guidance

Recommended loading level for MicroTex 950 is 4.0 – 6.0% by total formula weight to achieve the optimal effect. Lower levels may reduce gloss and improve edge coverage but will not form a texture. Too high of a level may restrict the flow of the coating and result in a discontinuous film.

MicroTex 950 is added at the premix stage and homogenized with the rest of the raw materials in the extruder. The material has a low bulk density, so the feed rate may need to be increased to achieve the same throughput as a powder formulation without the additive.

Because the texture formation is a physical phenomenon that occurs during curing, MicroTex 950 can be used in any powder coating binder chemistry and is independent of cure temperature.

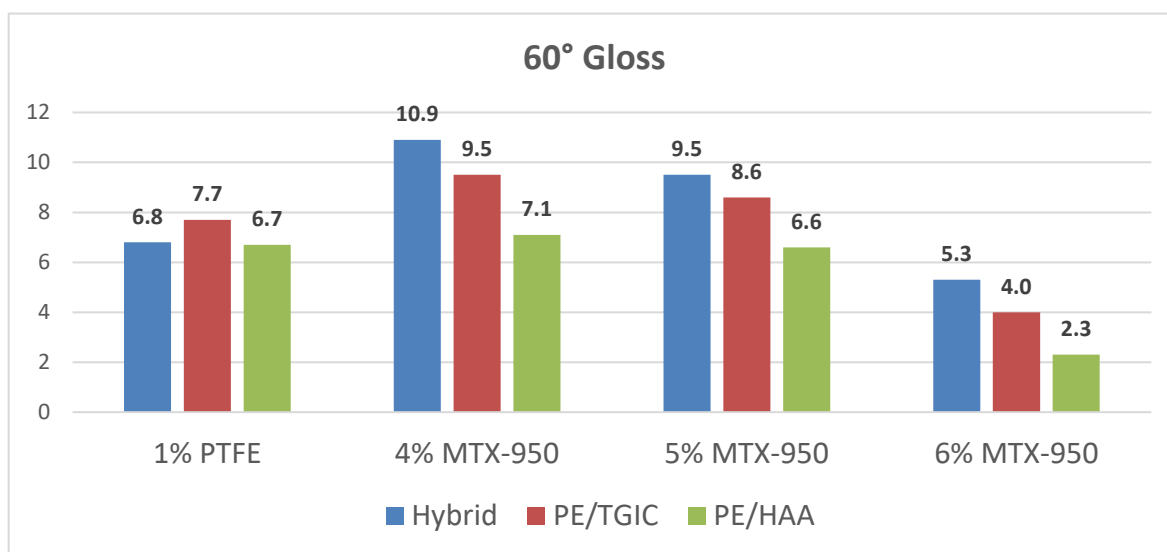
Sample Preparation

The following processing conditions were used for this study:

- Premix: 15 seconds, Vitamix blender
- Extrusion: APV 19mm co-rotating twin screw
 - 100°C/100°C set temperature, 500 RPM, 45-60% Torque
- Pulverization: Strand mill
- Sieve: 120 mesh
- Electrostatic Application: Nordson Encore
- Cure: Electric convection oven (15 minutes at 200°C (part temp))

Material	Description	1% PTFE	4% MTX-950	5% MTX-950	6% MTX-950
Resin/crosslinker (See Results table)	Stoichiometrically calculated for each resin chemistry				
Resiflow PL-200	Flow Agent	5.00	5.00	5.00	5.00
Benzoin	Degassing Agent	2.50	2.50	2.50	2.50
Raven 2500	Carbon Black	5.00	5.00	5.00	5.00
Portaryte B10	Barium Sulfate	145.00	130.00	125.00	120.00
Fluo 750TX	PTFE	5.00			
MicroTex 950	Modified polyolefin		20.00	25.00	30.00

Results



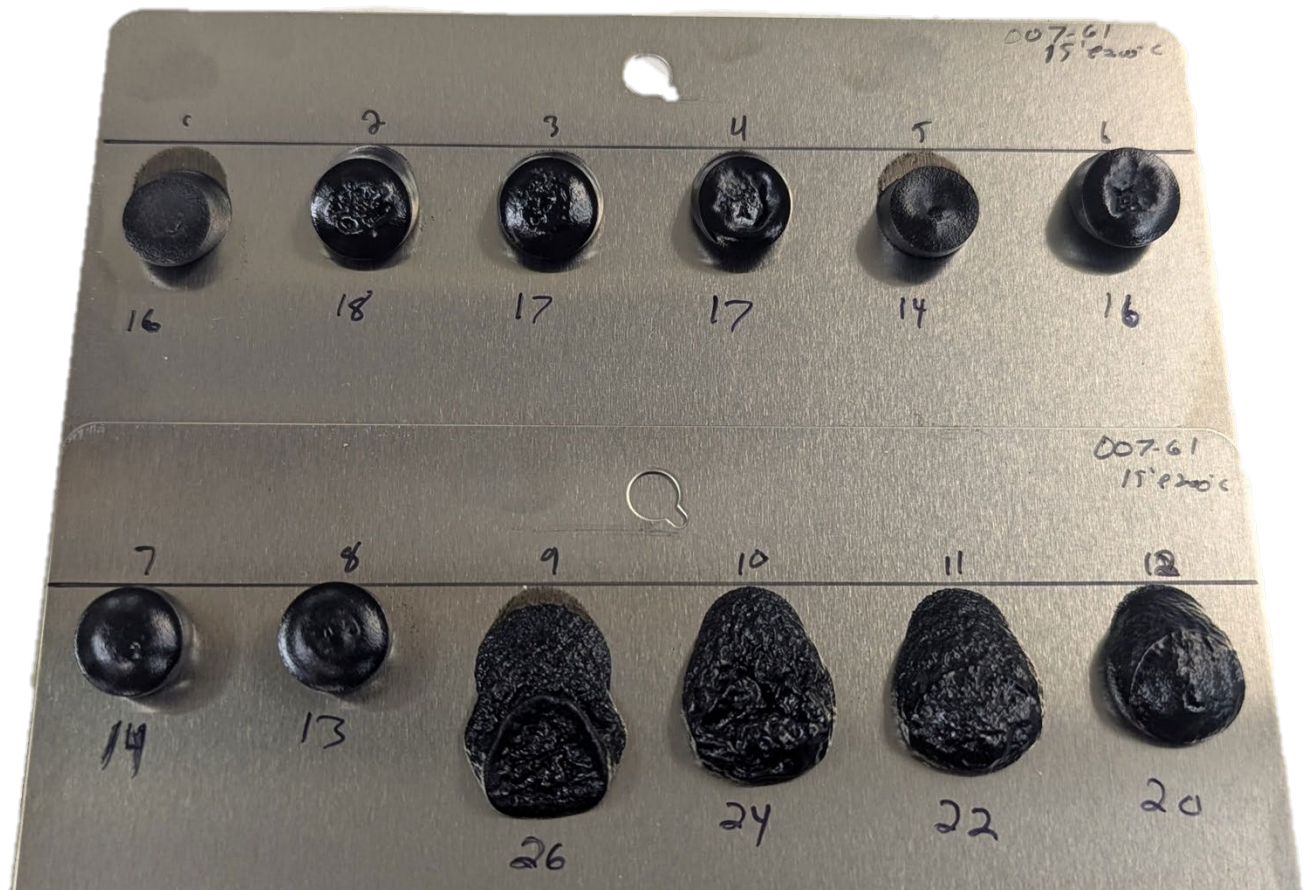
60° Gloss	1% PTFE	4% MTX-950	5% MTX-950	6% MTX-950
Hybrid	6.8	10.9	9.5	5.3
PE/TGIC	7.7	9.5	8.6	4.0
PE/HAA	6.7	7.1	6.6	2.3

Mechanical Properties

#	Chemistry	Description	Crosshatch Adhesion	Impact Resistance		Conical Mandrel	Pellet Flow
				Forward	Reverse		
1	Hybrid	1% PTFE	5B	100	160	Pass	16
2		4% MTX-950	5B	100	100	Pass	18
3		5% MTX-950	5B	160	100	Pass	17
4		6% MTX-950	5B	140	160	Pass	17
5	PE/TGIC	1% PTFE	5B	160	100	Pass	14
6		4% MTX-950	5B	160	160	Pass	16
7		5% MTX-950	5B	120	100	Pass	14
8		6% MTX-950	5B	80	60	Pass	13
9	PE/HAA	1% PTFE	5B	160	120	Pass	26
10		4% MTX-950	5B	120	60	Pass	24
11		5% MTX-950	5B	100	80	Pass	22
12		6% MTX-950	5B	80	60	Pass	20

Pellet Flow

Pellet flow is a test of melt/flow under low shear. In general, a longer pellet corresponds to a smoother coating. A shorter pellet has limited flow but better edge coverage.



Mechanical Properties

- MicroTex 950 did not influence adhesion to the substrate
- At a 6% loading level, the impact resistance of the coatings was slightly lower vs. the PTFE-containing control for the TGIC and HAA formulas, but was slightly higher in the hybrid system
- The mandrel bend flexibility of the coatings was not affected
- Pellet flows were similar in the hybrid and TGIC systems, and were reduced in the HAA system

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Guidance for White Formulations

Here are some comments on how to get best results in a white powder system:

1. The ratio by volume of the additive to the binder changes depending on the specific gravity of the fillers and pigments. The density of the barium sulfate used in our study is close to titanium dioxide, which would be used in a white formulation. However, there could be a big difference if the additive is in at a constant weight but you change 30% of the formula from calcium carbonate (specific gravity = 2.7) to TiO_2 (specific gravity = 4.2).
2. Texturing performance could be impacted when using a resin with low melt viscosity and low glass transition temperature. If MicroTex 950 is added to an otherwise very smooth and high gloss starting formula, the rheological effect will not be as pronounced.
3. The inert (filler and pigment) portion of the formula could have different oil absorption values. In this case, higher oil absorption would increase the texturing effect. This could also be affected by differences in particle size of the mineral fillers.